Accounting and Budgetary Systems for Engineers - INFO7225

MGEN

Accounting Information System (AIS) and Business Process Fundamentals (II)

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The Basics of Financial Accounting

Special topic 1: cash-basis vs. accrual-basis accounting

Special topic 2: merchandising transactions

Special topic 3: accounting for long-term assets

Summary: generally accepted acct. principles (GAAP)

The Basics of Financial Accounting:

Special Topic 2:

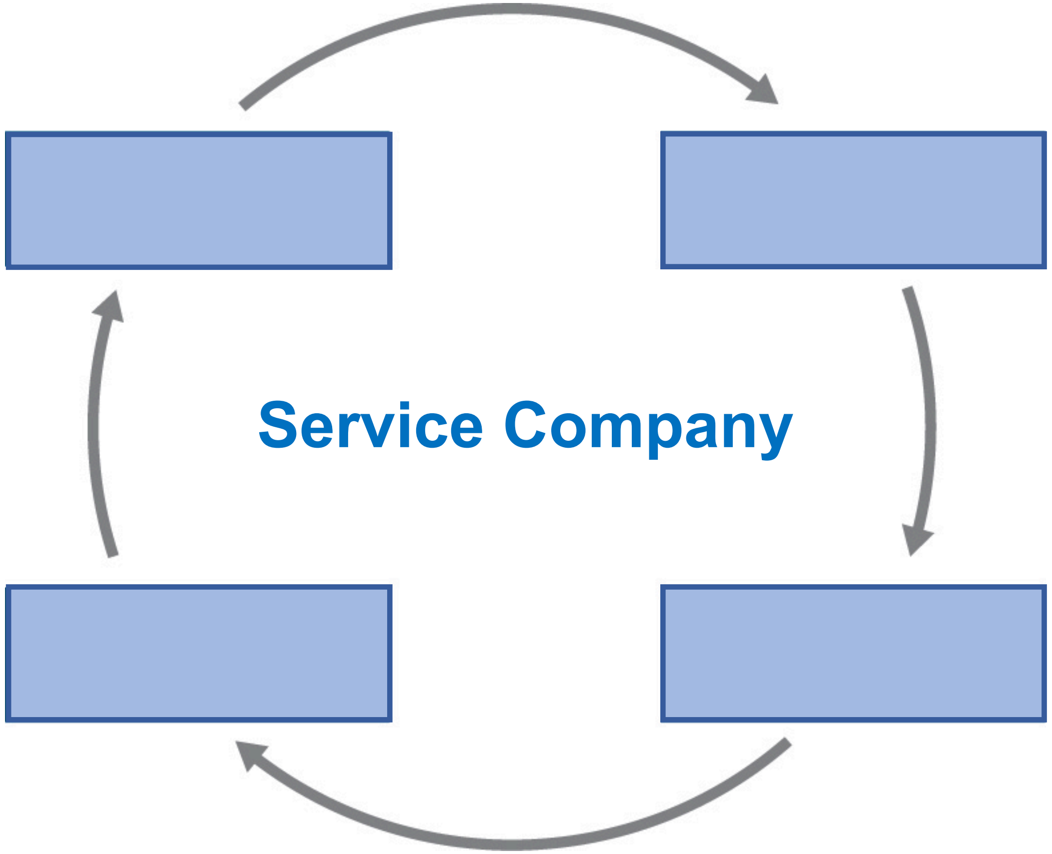
Merchandising Transactions

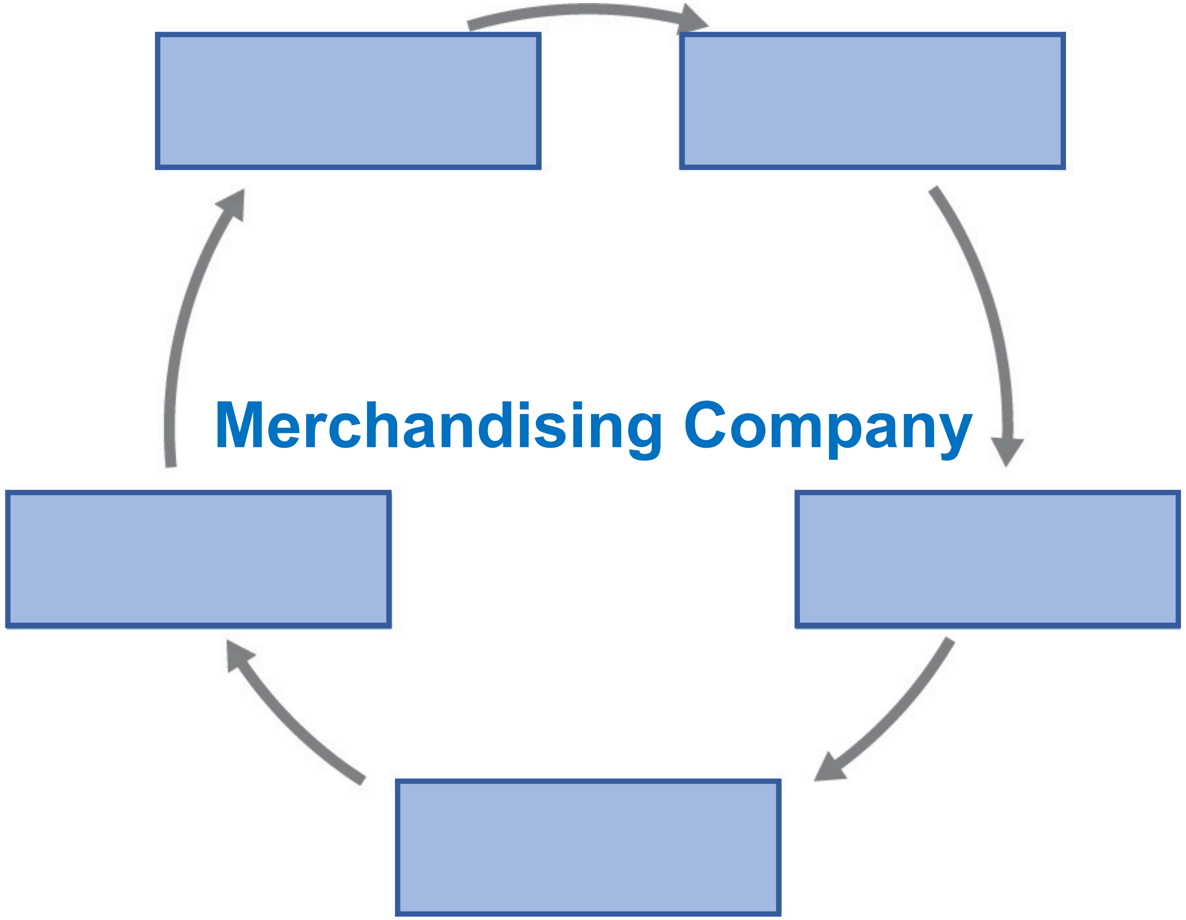
LEARNING OBJECTIVES

After completing this session, you should be able to

1. **Explain the difference between the two systems/methods that are used in merchandise inventory accounting: perpetual system and periodic system;**
2. **Describe the two cost flow assumptions commonly used in merchandise inventory accounting: first-in, first-out (FIFO) and last-in, first-out (LIFO);**
3. **Analyze revenues (sales), cost of goods sold (COGS), ending inventory, and gross profit using FIFO and LIFO.**

TYPICAL OPERATING CYCLE FOR A ___________:





Differences Between the Income Statements of a Service Company and a Merchandising Company

Walmart Inc. (WMT) Balance Sheet

All numbers in thousands

Breakdown	1/31/2022	1/31/2021	1/31/2020	1/31/2019
Total Assets	244,860,000	252,496,000	236,495,000	219,295,000
Current Assets	81,070,000	90,067,000	61,806,000	61,897,000
Cash, Cash Equivalents & S...	14,760,000	17,741,000	9,465,000	7,722,000
Receivables	8,280,000	6,516,000	6,284,000	6,283,000
Inventory	56,511,000	44,949,000	44,435,000	44,269,000
Prepaid Assets	1,519,000	20,861,000	1,622,000	3,623,000
Total non-current assets	163,790,000	162,429,000	174,689,000	157,398,000
Total Liabilities Net Minority Int...	152,969,000	164,965,000	154,943,000	139,661,000
Total Equity Gross Minority Inte...	91,891,000	87,531,000	81,552,000	79,634,000

https://finance.yahoo.com/quote/WMT/balance-sheet?p=WMT

Walmart Inc. (WMT)

Income Statement

All numbers in thousands

Breakdown	TTM	1/31/2022	1/31/2021	1/31/2020	1/31/2019
› Total Revenue	576,013,000	572,754,000	559,151,000	523,964,000	514,405,000
Cost of Revenue	432,575,000	429,000,000	420,315,000	394,605,000	385,301,000
Gross Profit	143,438,000	143,754,000	138,836,000	129,359,000	129,104,000
› Operating Expense	119,087,000	117,812,000	116,288,000	108,791,000	107,147,000
Operating Income	24,351,000	25,942,000	22,548,000	20,568,000	21,957,000
› Net Non Operating Interest Inc...	-1,719,000	-1,836,000	-2,194,000	-2,410,000	-2,129,000
› Other Income Expense	-4,879,000	-5,410,000	210,000	1,958,000	-8,368,000
Pretax Income	17,753,000	18,696,000	20,564,000	20,116,000	11,460,000

https://finance.yahoo.com/quote/WMT/balance-sheet?p=WMT

Walmart Inc. (WMT)

Cash Flow

All numbers in thousands

Breakdown	TTM	1/31/2022	1/31/2021	1/31/2020	1/31/2019
› Operating Cash Flow	17,565,000	24,181,000	36,074,000	25,255,000	27,753,000
› Investing Cash Flow	-16,423,000	-6,015,000	-10,071,000	-9,128,000	-24,036,000
› Financing Cash Flow	-12,114,000	-22,828,000	-16,117,000	-14,299,000	-2,537,000
› End Cash Position	11,922,000	14,834,000	17,788,000	9,515,000	7,756,000

https://finance.yahoo.com/quote/WMT/balance-sheet?p=WMT

Merchandise Inventory Accounting: Two Different Systems/Methods

	Perpetual System	Periodic System
Inventory purchase transaction:		
Sales transaction:		

Merchandising Company: Revenue, Cost of Goods Sold (COGS), and Ending Inventory Computation

Q1			Q2			Q3			Q4			*t*	
Beg. Inv.	Buy			Buy			Buy			Buy			
10	10			10			10			10		units	
$100	$110			$120			$130			$140		per unit	
		Sell			Sell			Sell			Sell		
		10			10			10			5	units	
		$150			$160			$170			$180	per unit	
Rev.	= $			= $			= $			= $			
Perpetual System:													
		COGS			COGS			COGS			COGS	Ending Inv.	
FIFO:													
LIFO:													

FIFO, first-in, first-out; LIFO, last-in, first-out

Merchandising Company: Revenue, Cost of Goods Sold (COGS), and Ending Inventory Computation

Q1			Q2			Q3			Q4				
Beg. Inv.	Buy			Buy			Buy			Buy		*t*	
10	10			10			10			10		units	
$100	$110			$120			$130			$140		per unit	
		Sell			Sell			Sell			Sell		
		10			10			10			5	units	
		$150			$160			$170			$180	per unit	
Rev.	= $			= $			= $			= $			
Perpetual System:													
		COGS			COGS			COGS			COGS	Ending Inv.	
FIFO:													
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Periodic System:													
		COGS			COGS			COGS			COGS	Ending Inv.	
FIFO:													
LIFO:													

FIFO, first-in, first-out; LIFO, last-in, first-out

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		$150			$160			$170			$180	per unit	
Rev.	= $			= $			= $			= $			
Periodic System:													
		COGS			COGS			COGS			COGS	Ending Inv.	
FIFO:													
LIFO:													

FIFO, first-in, first-out; LIFO, last-in, first-out

Perpetual System:		
	FIFO	**LIFO**
COGS		
Ending Inventory		

Periodic System:		
	FIFO	**LIFO**
Beginning Inventory		
Purchases		
Goods available for sale		
Ending Inventory*		
COGS		

*Ending inventory: from physical count at the end of an accounting period.

Cost flow assumption	**FIFO**	**LIFO**
Perpetual System		
Periodic System		

SUMMARY

- A perpetual inventory system inventory updates merchandise inventory and sales records constantly.
- A periodic inventory system only records updates to inventory and costs of sales at scheduled times throughout the year, not constantly. Merchandise Inventory and Cost of Goods Sold are updated at the end of a period.
- FIFO:
- LIFO:

ADDITIONAL NOTES

- IFRS does **not** permit the use of LIFO.
- Under GAAP, companies are free to choose among three ways to report COGS and ending inventory: FIFO, LIFO, or weighted avg.
  - Once the methods of costing are determined for the company, that methodology would typically be applied repeatedly over the remainder of the company’s history.
- Although the Internal Revenue Service (IRS) generally allows differing methods of accounting treatment for tax purposes than for financial statement purposes, an exception exists that
  - Prohibits the use of LIFO inventory costing on the company tax return unless LIFO is also used for financial reporting.

Questions?